

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electrical and Electronics Engineering
Academic Year: 2020 - 2021 (Even Semester)
Innovative Practices Description

Unit I IC Fabrication

Degree, Semester & Branch	: IV Semester B.E. EEE
Course Code & Title	: EE8451 Linear Integrated Circuits and its Applications
Name of the Faculty member	: Mrs. S. Jeyanthi
Name of the Topic	: Fabrication of FET
Name of the Innovative Practice	: Strip Sequence
Date & Duration	: 01.03.2021& 10 Minutes

Description:

Give students the steps in a process on strips of paper that are jumbled; ask them to work together to reconstruct the proper sequence.

Learning Outcome:

- The students will be able to visualize the concepts and organize information.

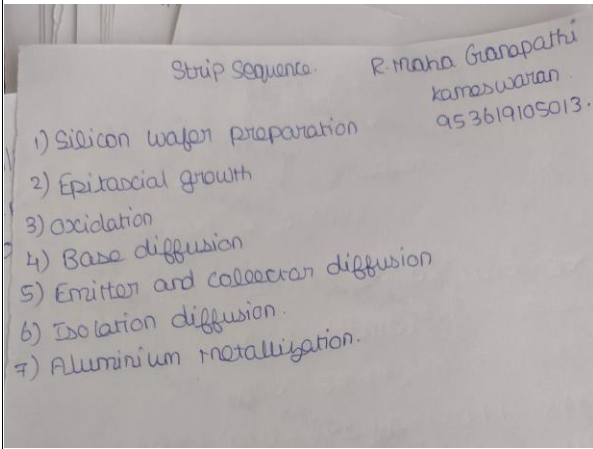
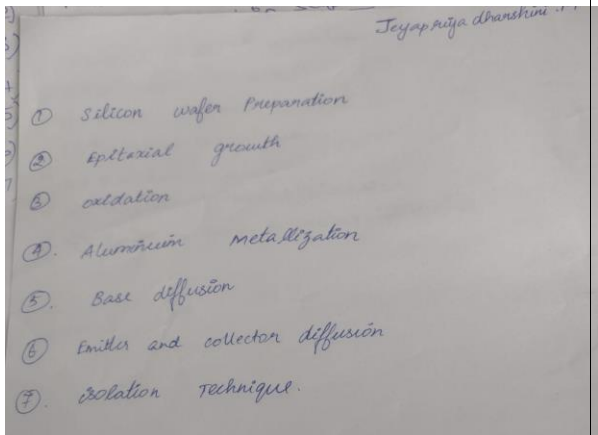
Justification for choosing the topic using Strip Sequence activity:

This activity helps students apply what they have learned through reading or didactic teaching. This approach can strengthen students' logical thinking processes and test their mental model of a process. The activity can be done in pairs or groups.

Implementation (Plan & Execution) with Proof:

Strip Sequence activities require students to actively engage in their learning, often by connecting their prior knowledge to new information. When creating a Strip Sequence, a student frequently interacts with a textbook, notes from class, an instructor, classmate, or study group.

In my subject Mini map was conducted for Unit-I IC Fabrication for the topic Fabrication of FET. The students are voluntarily did the arrangement. The sample copy of the Strip sequence was done by the students was shown below.

Innovative Teaching Method Execution	
Fabrication of FET– Strip sequence	
 Strip Sequence. R. Maha Ganapathi Kameswaran. 953619105013. 1) Silicon wafer preparation 2) Epitaxial growth 3) oxidation 4) Base diffusion 5) Emitter and collector diffusion 6) Isolation diffusion. 7) Aluminium metallization.	 Jayapriya dhanakumari ① silicon wafer preparation ② epitaxial growth ③ oxidation ④ Aluminium metallization ⑤ Base diffusion ⑥ emitter and collector diffusion ⑦ isolation technique.

Reflective Critique:

Benefits:

The benefits of Strip Sequence are a great way for students to make notes on all of the information they receive. It made the students the general steps involved in the process and make the students to implement the procedure for the particular topic.

Challenges:

Normally teachers will give longer explanations in the notes section of the topic. The students are made into groups and to arrange the strip of fabrication process for the topic of fabrication of FET. I planned the activity for 10 minutes only. But in real scenario it takes 15 minutes to complete the whole topics in Unit-I.

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2	1	1	-	-	1	2	1	-	1

Reference:

1. D. Roy Choudhary, Sheil B. Jani, 'Linear Integrated Circuits', II Edition, NewAge, 2003.

CO1: After completion of the course, it is expected that students will be able to describe the IC fabrication procedure.

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Department of Electrical and Electronics Engineering
Academic Year: 2020 - 2021 (Even Semester)
Innovative Practices Description

Unit II Characteristics of Op-Amp

Degree, Semester & Branch	: IV Semester B.E. EEE
Course Code & Title	: EE8451 Linear Integrated Circuits and its Applications
Name of the Faculty member	: Mrs. S. Jeyanthi
Name of the Topic	: Basic Application of op-amp – Inverting amplifier & Non-inverting amplifier
Name of the Innovative Practice	: Demonstrations
Date & Duration	: 11.03.2021 & 15 Minutes

Learning Outcome:

- The students will be able to visualize the concepts and able to understand the concept.

Justification for choosing the topic using demonstration:

After teaching the concept, I thought of conducting this activity for making the students to give the knowledge about the characteristics of Basic Application of op-amp – Inverting amplifier & Non-inverting amplifier which enhance the learning level and as a teacher I can judge the understanding level of the students.

Implementation (Plan & Execution) with Proof:

Strip Sequence activities require students to actively engage in their learning, often by connecting their prior knowledge to new information. When creating a Strip Sequence, a student frequently interacts with a textbook, notes from class, an instructor, classmate, or study group.

In my subject Mini map was conducted for Unit II Characteristics of Op-Amp for the topic Basic Application of op-amp – Inverting amplifier & Non-inverting amplifier. The students are voluntarily did the arrangement. The sample copy of the Strip sequence was done by the students was shown below.

Innovative Teaching Method Execution	
Basic Application of op-amp – Inverting amplifier & Non-inverting amplifier - Demonstrations	
	

Reflective Critique:

1. Pre-implementation Reflection :

- **Benefits:**

- Students can able to attend the question even in the questions are in indirect form.
- Students can able to explain the concepts in examination without any confusion.

- **Challenges:**

- In the class mostly girls hesitate to answer the questions.
- Time utilization for conducting activity.

2. Post-implementation Reflection :

- **Benefits:**

- Students understood the concept which was reflected from their answers for the questions I have asked during discussion session.

- **Challenges:**

- Slow learners were not able to answer the question.

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2	1	1	-	-	1	2	1	-	1

Reference:

1. D. Roy Choudhary, Sheil B. Jani, 'Linear Integrated Circuits', II Edition, NewAge, 2003.

CO1: After completion of the course, it is expected that students will be able to determine the AC and DC characteristics of op-amps and signal analysis using op-amp based circuits.